Supplementary Material for "A Semantic Mutation Metric for Metamorphic-Relation Adequacy in Scientific Computing Programs"

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CCS Concepts: • **Software and its engineering** → **Software testing and debugging**; *Software verification and validation.*

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A A. Notation and Operator Catalogue

A.1 A.1 Complete notation table

Experimental objects.

- PUT set $I = \{\text{A1,A2,A3, B1,B2,B3, C1,C2,C3, D1,D2,D3}\}$, $|I| = 12$.
- PUT: S_i $(i \in I)$.
- Class mapping: $\text{cls} : I \to \{A, B, C, D\}$ (open, extensible).
- Valid input distribution: D_S, sampling $X_{K_{\text{eq}}} \sim D_S$.

Metamorphic testing side.

- Meta-Pattern (MP): $\text{MP} = \{\text{MP}_1, \ldots, \text{MP}_5\}$, $k \in \{1, \ldots, 5\}$ (open, extensible). MP_1 Conservation, MP_2 Monotonicity, MP_3 Convergence, MP_4 Trajectory, MP_5 Partial-order.
- Metamorphic Relation: $\text{MR}_{i,k}$ (from the shared infrastructure); $mr = (r, R) \in \text{MR}_{i,k}$.
- Automated Verification Pipeline (AVP): $\text{AVP} : \text{Programs} \times \text{MR}_{\text{universe}} \times \mathbb{R}^+ \to \{\text{pass}, \text{fail}\}$; $\text{AVP}(s, mr, \varepsilon^k_{\text{AVP}})$ invokes the verification method per MP_k.

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Mutation testing side.

- Mutation operator family: $\mathrm{MUT} = \{\mathrm{mut}_1, \dots, \mathrm{mut}_5\}$, $j \in \{1, \dots, 5\}$ (open, extensible); $\mathrm{mut}_1 = \mathrm{mut}_C$, $\mathrm{mut}_2 = \mathrm{mut}_M$, $\mathrm{mut}_3 = \mathrm{mut}_G$, $\mathrm{mut}_4 = \mathrm{mut}_T$, $\mathrm{mut}_5 = \mathrm{mut}_F$.
- Signature: $\mathrm{mut}_j : \mathrm{Programs} \to 2^{\mathrm{Programs}}$.
- Mutant: $s' \in \mathrm{mut}_j(S_i)$.
- Alignment: $\mathrm{align}(j) = j$.

Three-state decomposition (fixed).

$$\mathrm{mut}_j(S_i) = \mathrm{equiv}_{i,k,j} \cup \mathrm{killed}_{i,k,j} \cup \mathrm{survive}_{i,k,j},$$

with the three sets disjoint, mutually exclusive, and exhaustive.

- *equiv determination* (dual conditions): (E1) AVP-coherent: $\forall\, mr \in \mathrm{MR}_{i,k} : \mathrm{AVP}(S_i, mr) = \mathrm{AVP}(s', mr)$; (E2) Output-equiv: $\forall x \in X_{K_{\mathrm{eq}}} \sim D_S : \|S_i(x) - s'(x)\| \le \varepsilon_{\mathrm{eq}}$.
- *killed determination*: $\mathrm{killed}(s', \mathrm{MR}_{i,k}) \Leftrightarrow \exists mr \in \mathrm{MR}_{i,k} : \mathrm{AVP}(S_i, mr) = \mathrm{pass} \wedge \mathrm{AVP}(s', mr) = \mathrm{fail}$.

Metric (fixed classical structure).

$$\begin{aligned}\mathrm{SMS}_{i,k,j} &:= \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{mut}_j(S_i)| - |\mathrm{equiv}_{i,k,j}|} \\ &= \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{killed}_{i,k,j}| + |\mathrm{survive}_{i,k,j}|} \in [0, 1].\end{aligned}$$

LRCA engineering attribution layer (descriptive, not in SMS)..

- Likely root-cause inventory: $C = \{C_1, \dots, C_5\}$ (open, extensible). C1 True semantic failure; C2 Tolerance perturbation; C3 OOD; C4 Statistical-assumption violation; C5 Mutator artefact.
- LRCA output: each $s' \in$ killed annotated with $\texttt{root\_cause}(s') \in C$.
- Descriptive quantities: $\texttt{C1\_share}_{i,k,j}$, $\texttt{suspect\_share}_{i,k,j}$.

Slicing and cross-class.

- Slicing: aligned $j = k$; cross $j \neq k$.
- Cross-class: $\Delta\mathrm{SMS}_c$, $c \in \{A, B, C, D\}$; $\mathrm{CV}(\Delta\mathrm{SMS}) := \mathrm{std}(\Delta\mathrm{SMS}_c)/|\mathrm{mean}(\Delta\mathrm{SMS}_c)|$.

Tolerance and sampling.

- Tolerance: $\varepsilon^k_{\mathrm{AVP}}$ (MP-dependent), $\varepsilon_{\mathrm{eq}}$ (equivalence output tolerance).
- Sampling: $K_{\mathrm{eq}} = 1000$ (E2 sample size), $N = 20$ (statistical replicates).

The notation skeleton (11 core concepts + three-state decomposition + SMS formula + AVP interface) is **fixed**; the *content* of MUT, MP, C, and cls is open to extension.

A.2 A.2 LRCA decision tree and three-layer diagnostic protocol

Five likely root causes (open):

Table 1. LRCA decision tree and three-layer diagnostic protocol.

Code	Meaning
C1	True semantic failure (what SMS seeks)
C2	Numerical tolerance perturbation
C3	Out-of-distribution (OOD)
C4	Statistical-assumption violation
C5	Mutator artefact

Three-layer diagnosis with L0 artefact pre-scan:

- **L0 artefact pre-scan.** Double-blind review (§3.9.4 dual-LLM cross-source + 20% manual sampling).
- **L1 tolerance robustness.** $N = 20$ replicates; fail ratio ≥ 0.80 considered stable.
- **L2 OOD triage.** For C/D classes, distinguish valid D_S^{valid} from OOD.
- **L3 statistical baseline.** For Wilcoxon/DTW, IID/stationarity pre-check on PUT samples.

Decision tree:

For each $s' \in \text{killed}_{i,k,j}$:

- **L1.** fail ratio $< 0.80 \Rightarrow$ C2; otherwise $\Rightarrow$ L2.
- **L2** (C/D classes). fail only in OOD $\Rightarrow$ C3; otherwise $\Rightarrow$ L3.
- **L3** (B/D classes + Wilcoxon/DTW). assumption violation $\Rightarrow$ C4; otherwise $\Rightarrow$ artefact recheck.
- **Artefact recheck.** artefact evidence $\Rightarrow$ C5; otherwise $\Rightarrow$ C1.

Multi-label priority: C5 > C4 > C3 > C2 > C1.

LRCA threshold calibration (9-grid scan; `lrca_calibration.json`). Best ood_band=0.02 with any tolerance_multiplier (3.0 / 10.0 / 30.0) gives mean_C1_share 0.200 / H4 12/60 (20.0%); ood_band=0.05 (default) or 0.10 gives 0.164 / 10/60 (16.7%). `tolerance_multiplier` has zero impact (L1 tolerance rarely triggered); `ood_band` is the only discriminator. Calibration ceiling 12/60 = 20% remains far below 80%, H4 unattainable on this dataset, not a calibration issue (inherent SNR of LLM-mutant pools).

A.3 A.3 AVP protocol specification and equivalence-judgement trade-off

$$\text{AVP} : \text{Programs} \times \text{MR}_{\text{universe}} \times \mathbb{R}^{+} \to \{\text{pass}, \text{fail}\}.$$

Per-MP verification: MP_1 tolerance equality $|\text{LHS} - \text{RHS}| \leq \varepsilon$; MP_2 / MP_5 Wilcoxon signed-rank $\alpha = 0.05$; MP_3 convergence-order + asymptotic residual ratio; MP_4 DTW distance threshold ε_{DTW}. AVP version = upstream infrastructure commit hash; our package embeds AVP source.

Three-candidate equivalence trade-off.

Table 2. AVP protocol specification and equivalence-judgement trade-off.

Determination	False-positive	False-negative	SMS bias
E1 alone	numerical coincidence + AVP-coherent	E1 false → non-equiv	Low (smaller denominator)

E2 alone	output match, MR differs (rare)	K_eq miss, full-space equal	High
E1 ∧ E2	both mis-pass simultaneously	E1 ∨ E2 false → non-equiv	Slightly high (conservative)

E1 alone is fooled by insufficient AVP coverage; E2 alone by numerical coincidence on K_{eq}. E1 ∧ E2 is the conservative complete instantiation; under L_{equiv} it reduces almost-everywhere to classical bitwise equivalence (Lemma 9.1). Counter-examples: E2 passes but E1 does not (rare; mutant coincides at K_eq but deviates at MR trigger points → non-equivalent); E1 passes but E2 does not (common; consistent within MR but numerical drift > ε_{eq} → non-equivalent).

B B. Experimental Subjects and Mutation-Operator Specialisations

B.1 B.1 PUT selection rationale

(a) Library-stack coverage. numpy 2.4.4 (A1-B3 linear algebra / arrays), scipy 1.17.1 (A1 integrate, A2 linalg, B2 stats), scikit-learn 1.8.0 (C1-D3 surrogate / ML), Python scientific-computing's de facto foundation stack (PyPI top-50, 2026-04). Uncovered: GPU / distributed (JAX, CuPy, dask), domain-specific (BioPython, Astropy, RDKit). Left to future work (R5).

(b) Mathematical-structure coverage. ODE/PDE (A1, A3), direct linear algebra (A2), Bayesian analytic + MCMC + Monte Carlo (B1-B3), kernel + orthogonal polynomial + NN surrogate (C1-C3), convex optimisation + backprop + max-likelihood (D1-D3), covers 8 of 12 chapters in Numerical Recipes [2]. Uncovered: advanced PDE solvers (FEM, FV, spectral); FFT; interior-point/trust-region optimisation; symbolic/CAS. Left to future work.

(c) Comparison with benchmarks. DeepCrime (Humbatova et al. 2021): semantic class-D topical overlap on ML kernels. Defects4J (Just et al. 2014): no overlap, only mutation-as-fault-proxy reference. mutmut / cosmic-ray demos: general Python; the semantic operators extend the scientific-computing focus.

(d) Scale vs representativeness. Each PUT 50-400 LOC; signature standardised to `program(x: float) → float`. Substantively smaller than industrial code (typically 1-10 KLOC).

Limitation. The `float → float` signature is a substantive constraint, not engineering trade-off. Industrial inputs are typically high-dimensional (CFD grids, MD particles, FEM matrices); scalarised PUTs upper-bound mutant semantic complexity and may systematically under-estimate SMS and cross-class differences on industrial PUTs (declared §6 R5; future work validates on industrial-grade PUTs).

B.2 B.1.5 Systematic vs incidental

Syntactic tools may *occasionally* hit §3.2 (a) (cross-function-boundary) or (c) (algorithm class), but occasionality undermines two engineering functions of semantic mutation. **(a) Deepening source-code understanding.** Designing OS `np.linalg.det(M) → np.sum(np.diag(M))` requires knowing the two are equivalent on diagonal matrices but not on general matrices (`det = ∏ eigvals` vs `sum(diag) = trace = ∑ eigvals`); syntactic tool AST traversal does not require this understanding even when similar replacements appear. **(b) Revealing deep faults.** Domain-semantic errors (physical-constant, unit conversion, boundary conditions, hyperparameter semantics, numerical-method order) are not AST-local; syntactic mutator design goals (operator typos, off-by-one, negation flips) have hit probability for domain errors much smaller than systematic semantic-mutator triggering, and lack repeatability. **Conclusion.** Syntactic tools occasionally producing mutants satisfying (a)(b)(c) is stochastic byproduct, not repeatable, not carrying engineering value. Systematic semantic mutation requires (a)(b)(c) to be design intent.

B.3 B.2 Per-class operator specialisations

Table 3. Per-class operator specialisations.

Operator	Representative specialisations across PUTs
mut_C Conservation-breaking	A1 Lorenz: add $\varepsilon_{\text{drift}}$ to RHS (slow Hamiltonian drift); A2 LU: decomposition omits k+1-th multiplier (breaks det conservation); B1 Beta-Bin: posterior omits normalisation; C1 GPR: covariance omits positive-definite diagonal term; D1 MLP: backprop omits one gradient term.
mut_M Monotonicity-breaking	A3 FDM: Δt coefficient occasionally negative; B2 MCMC: acceptance min(1, r) → min(0.95, r); C2 PCE: high-order coefficient sort inserts inversion; D2 SVM: decision-function sign flips near boundary.
mut_G Convergence-breaking	A1 Lorenz: RK4 → 1.5-order hybrid; A3 FDM: 2nd-order difference → 1st-order; B3 MC: doubling sample size does not 1/N-reduce variance; C3 NN-Surr: training-epoch truncation.
mut_T Trajectory-distorting	A1 Lorenz: state-vector y/z swap; B2 MCMC: insert independent-sampling segment; C3 NN-Surr: training-target slow phase shift; D1 MLP: hidden-layer periodic-mask activation.
mut_F Fidelity-order-breaking	A2 LU: partial pivoting degrades to no pivoting; C1 GPR: length-scale switches to coarse prior; C2 PCE: high-order term randomly retains low-order; D3 LR: regularisation occasionally large.

Per-class HP / OS / TF substitution rules (illustrative). **HP** on class a: tolerance / max_iter; on class c: GPR `noise_level` / `length_scale`; on class d: MLP `hidden_dim` / dropout. **OS** on class a: numerical-linalg API swaps (`det` ↔ `sum(diag)`); on class b: probability-distribution sampling swaps; on class c: surrogate-class swaps (GPR ↔ RBF ↔ NN). **TF** on class a: integration-order changes (RK4 → Euler); on class b: MC estimator changes.

Operator-level cross-table with cosmic-ray defaults. The 12 cosmic-ray default operators are AST-local: `NumberReplacer` (Num/Constant, CE partial, △ literal values only); `ReplaceArithmeticOperator` (BinOp); `ReplaceComparisonOperator` (Compare); `ReplaceLogicalOperator` (BoolOp); `ReplaceUnaryOperator` (UnaryOp); `ReplaceTrueFalse` (NameConstant, CE keyword, △ Bool literal); `BreakContinueReplacer` (Break/Continue);

`RemoveDecorator`; `RemoveExceptHandler`; `ZeroIterationForLoop` (For); `ReplaceIfBlock` (If); `MutateSubscript`. None correspond to OS / HP / TF / SI:

Table 4. Per-class operator specialisations.

Semantic class	Tool support	Coverage
OS API replacement	None	△ 88.33% disjoint (incidental low-complexity hits)
HP	None	× tool inexpressible (0/72)
TF numerical-method order	None	× tool inexpressible (0/54)
SI / CF structural injection	None	× tool inexpressible (0/33)

Operator-level conclusion. All 12 default classes remain AST-local (BinOp / Compare / BoolOp / UnaryOp / NameConstant / Subscript / If / For / Break / Continue / decorator / except). Categorically, no entry recognises sklearn / scipy hyperparameter semantics (HP), numerical-method order (TF), or control-flow intent (SI / CF). For OS, low-complexity sub-expressions can be incidentally hit by BinOp; empirical 11.67% (§3.5) is consistent with 88.33% AST-disjointness lower bound.

B.4 B.3 cosmic-ray a1 single-PUT pilot

Before the 12-PUT generalisation (§3.5 main), a single-PUT lightweight empirical used `scripts/run_cosmic_ray_a1.sh` on a1 (file 934 B, AST-bound). Outcome: mutants_generated ~tens; ≥ 90% belong to BinOp / Compare / 12 default classes; small killed-by-`test_a1.py` ratio (PUT unit tests are output-shape/type sanity checks; most cosmic-ray mutants not killed), an empirical manifestation of the §3.2.6 thesis that syntactic-tool mutants are not aligned with MR-violation detection. The 12-PUT generalisation (§3.5) supersedes this pilot but corroborates the same conclusion at scale.

B.5 B.4 Expected LRCA risk profile

PUT-class × LRCA-layer risk weights. A numeric: C2 dominant (⋆⋆⋆ on L1 tolerance). B probabilistic: C4 dominant (⋆ ⋆ ⋆ on L3). C surrogate: C3 dominant (⋆ ⋆ ⋆ on L2 OOD; ⋆⋆ tolerance). D ML: C3 + C4 mixed (⋆ ⋆ ⋆ L2; ⋆⋆ L3).

Operator-PUT root-cause hotspots (expected). A: mut_C/G/F dominantly C2 (numerical tolerance), mut_M/T C1 (true semantic). B: dominantly C4 (statistical-assumption violations). C: mut_C/M dominantly C2/C1 on c1/c2, mut_T/F dominantly C3 on c1/c2; c3 (NN-Surr) row dominantly C5 (artefact). D: d1 row dominantly C5 (training-noise artefact); d2/d3 dominantly C1 / C3 mixed.

Expected suspect_share thresholds. A: 0.10-0.20 (acceptance ≤ 0.25); B: 0.20-0.35 (≤ 0.40); C: 0.20-0.30 (≤ 0.35); D: 0.25-0.40 (≤ 0.45). 60-cell average: ≤ 0.20 (= H4; acceptance ≤ 0.25).

B.6 B.5 Engineering-significance mapping

j = k aligned diagonal is the H2 threshold-test slice; j ≠ k off-diagonal is control; 6 vacant cells (∘, no MR exercised) are not formally adjudicated and are repurposed in §5.2 as descriptive evidence for R_sem / R_kill decoupling. The v3 → v4 micro-change (−0.009) attributes to LLM-source diversity under a fixed prompt; this is the source-axis null contrast underpinning §4.3.

B.7 B.6 Mutmut vs cosmic-ray default-operator overlap

Reviewer-requested manual operator-class cross-reference between cosmic-ray's 13 default operators (cosmic-ray 8.4.6, `cosmic_ray.operators` package) and mutmut's 14 default operators (mutmut current `main`, `src/mutmut/mutation/mutators.py`). Both default-operator sets are first-order AST-local replacements; neither implements a cross-function-boundary, domain-knowledge-dependent, or algorithmic-class-changing mutation (the three §3.2 necessary conditions for semantic mutation).

Table 5. Mutmut vs cosmic-ray default-operator overlap.

Operator class	cosmic-ray default	mutmut default	Reaches §3.2 (a/b/c)?	Semantic-class reachability
Numeric literal ±1	`number_replacer`	`operator_number`	(a) no, (b) no, (c) no	partial CE only (e.g. `_RHO=28.0` → `27.5`)
Binary arithmetic swap (+, −, ×, ÷)	`binary_operator_replacement`	`operator_swap_op` (binary subset)	(a) no, (b) no, (c) no	partial OS only (incidental)
Comparison swap (<, ≤, >, ≥, ==, !=)	`comparison_operator_replacement`	`operator_swap_op` (compare subset)	(a) no, (b) no, (c) no	none
Boolean swap (and ↔ or)	`boolean_replacer`	`operator_swap_op` (bool subset)	(a) no, (b) no, (c) no	none
Unary remove / swap	`unary_operator_replacement`	`operator_remove_unary_ops`, `operator_swap_op`	(a) no, (b) no, (c) no	none
Keyword swap (True/False, etc.)	`keyword_replacer`	`operator_keywords`, `operator_name`	(a) no, (b) no, (c) no	none
break ↔ continue / return	`break_continue`	`operator_keywords` (subset)	(a) no, (b) no, (c) no	none
Exception class swap	`exception_replacer`	(none)	(a) no, (b) no, (c) no	none
Decorator removal	`remove_decorator`	(none)	(a) no, (b) no, (c) no	none
Variable-binding insert / replace	`variable_inserter`, `variable_replacer`	`operator_arg_removal`, `operator_assignment`	(a) no, (b) no, (c) no	none
no_op insertion	`no_op`	(none)	(a) no, (b) no, (c) no	none
Zero-iteration `for` loop	`zero_iteration_for_loop`	(none)	(a) no, (b) no, (c) no	none
String content tweak (XX prefix, case flip)	(none)	`operator_string`	(a) no, (b) no, (c) no	none
Lambda body → `None`/`0`	(none)	`operator_lambda`	(a) no, (b) no, (c) no	none
Dict kwarg name (XX prefix)	(none)	`operator_dict_arguments`	(a) no, (b) no, (c) no	none
Sym./asym. string-method swap	(none)	`operator_symmetric_string_methods_swap`, `operator_unsymmetrical_string_methods_swap`	(a) no, (b) no, (c) no	none
Augmented → simple assignment	(none)	`operator_augmented_assignment`	(a) no, (b) no, (c) no	none
`match` case removal	(none)	`operator_match`	(a) no, (b) no, (c) no	none

Aggregate. Cosmic-ray ∩ mutmut: ~9 of 13 cosmic-ray operators have a near-equivalent in mutmut (numeric, binary, compare, boolean, unary, keyword, break-continue, variable, plus partial overlap on string-prefix vs `XX`); 4 cosmic-ray operators are unique (exception, decorator, no_op, zero-iteration-for); 6 mutmut operators are unique (string content, lambda, dict-kwarg, string-method swap, aug-assignment, match-case). The union is approximately 21 distinct first-order AST-local operator classes. None of the 21 can:

- (a) Cross a function-call or module-import boundary (e.g. replace `det(M)` with `sum(np.diag(M))` requires knowing the equivalence on diagonal matrices, outside any default-operator scope);
- (b) Depend on domain knowledge for legality (e.g. a hyperparameter swap that knows `noise_level=1e-4` is the load-bearing knob in a Gaussian Process Regression PUT requires PUT-class awareness);
- (c) Change the algorithmic class (e.g. swap `polynomial → spline basis` rewrites the surrogate's mathematical structure).

The mutmut / cosmic-ray null intersection on §3.5's HP/SI/TF classes (zero overlap) therefore reflects a **structural property of first-order AST-local mutation** rather than a tool-specific limitation, supporting the §3.6 preventive-defence framing.

C C. Experimental Procedure Details

C.1 C.1 Cross-source protocol, full specification

Two-stage ablation.

Table 6. Cross-source protocol, full specification.

Version	Mutant pool	c-class primary MP	Use
v3	Single-source (Claude Opus 4.6)	MP5 (pre-registered)	H2 baseline
v4	**Cross-source** (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat)	MP5 (held at pre-registered)	Isolate LLM-source diversity

Protocol (`scripts/cross_source_campaign.py`). (a) For each (PUT, operator), Claude / GPT / DeepSeek run K = 3 trials with identical prompt (§3.9.2), temperature 0.7; `source_tag` propagated to filenames. (b) **V1-V4 mechanical validation** (`src/p2/mutators/validation.py`): V1 syntax (`ast.parse`), V2 executability, V3 non-triviality (`|y_mutant - y_original| > 1e-6`), V4 signature consistency. (c) **DeepSeek model:** `deepseek-chat` (113 tokens/call, 1.8 s), not `deepseek-v4-pro` (340 tokens/call, 11 s), quality-equivalent on a2_OS1 dry-run. (d) **Pool capacity:** 37 ops × 3 sources × 3 trials = 333 attempts; V1-V4 pass rate 89.5%; 298 confirmed mutants, of which 292 enter the v4 pool after final-stage rejection of duplicates and QA failures. Sources contribute nearly equally: Claude 101 / GPT 98 / DeepSeek 99 (Phase A engineering finding). (e) **Sampling** (`scripts/build_pools.py`, POOL_VERSION = v4): per-PUT 30 max, measured mean 24.3, range 10-30. c1 (GPR) has only 10 because c1_HP1 / c1_CE1 have V1-V4 near-zero pass (WhiteKernel noise_level 1e-4 → 1e-1 perturbation has minimal output impact, triggers V3 non-trivial failure, itself §5.2 evidence).

Protocol-asymmetry (R13). v4 does not invoke reviewer LLM (cost / speed priority; a follow-up study will rerun dual-blind on the v4 grid, ~$5-8 USD); v3 used the original Phase-1 dual-blind (Claude gen + GPT-5.4 review + DeepSeek arbitration). A fraction of the v3 → v4 source-axis shift may be slight v4 quality decline rather than LLM-source diversity.

C.2 C.1.1 Differential prompt protocol

The v3 → v4 micro-change of −0.009 may reflect "three LLMs under identical prompt" rather than "true upper bound of LLM diversity". Separation requires *one tailored differential prompt per LLM* (skeleton: `scripts/run_differential_prompt.py`).

Design (2×3 factorial, within-(PUT, operator)). Factor A: prompt template, 3 levels: **V_canonical** (control, identical to v4), **V_persona** (per-LLM identity: Claude "numerical-analysis editor"; GPT "scientific-software refactorer"; DeepSeek "library-API substitution specialist"), **V_cot** (Claude `<thinking>` tags; GPT "step by step"; DeepSeek stepped-reasoning). Factor B, LLM source (Claude / GPT-5.4 / DeepSeek chat). K = 3 per cell; total 37 × 3 × 3 × 3 = **999 trials**.

Exit criteria. If max(delta) − min(delta) < 0.05: the v3 → v4 −0.009 source-axis null is robust. If ≥ 0.05 with prompt variant pushing delta past 0.474: H2 revised from "not met" to "prompt-conditional met". If ≥ 0.05 but all delta < 0.474: prompt sensitivity exists but H2 unchanged.

Resources. API ~$18-30 USD; wall 3-4 h at concurrency 4; V1-V4 only, no reviewer LLM.

C.3 C.2 Manual pilot lineage, LLM-client configuration, and L0 prescreen

The original §3.9 protocol used 60% multi-LLM consensus (Claude Opus 4.6 + GPT-5.4 + DeepSeek chat unanimous) plus 40% manual injection by scientific-computing researchers. Each (mut_j, S_i) pair carried one prompt template containing PUT source, semantic intent of mut_j, prohibition against revealing specific MRs (avoids prompt leakage), and output requirements (syntactically correct, executable, single-point modification, diff < 10 lines). Reproducibility parameters: temperature 0.3 in manual-pilot stage (v4 cross-source uses 0.7 per §C.1); fixed seed; 5 candidates per prompt with 2-3 retained.

Double-blind review (Scheme C). Generator LLM-G (Claude Opus) and reviewer LLM-R (GPT-5.4) are cross-source with strict role separation. LLM-R sees only PUT original + mutant code, unaware of mut_j category / MR content / generator identity, outputting a triple (syntactic, executable, semantic-fault-injected). Double-confirmed mutants enter pool; inconsistencies enter manual arbitration (≤ 10%). From the double-confirmed pool, 20% is sampled for manual review; manual-vs-LLM-R inconsistency > 10% triggers full manual downgrade. Same-source bias mitigation stratifies LLM-R by PUT class. Prompt-injection mitigation disables RAG / web search on the API path.

L0 pool prescreen. Each mutant passes static syntax (linters / type checkers), a unit self-test (simple inputs produce finite output), and double-blind sign-off. Target: 30-50 candidates per cell → 10-15 retained after prescreen.

C.4 C.4 LRCA three-layer execution

Decision tree, multi-label priority (C5 > C4 > C3 > C2 > C1), and output (`C1_share`, `suspect_share`) are stated in §A.2; the full pseudocode is reproduced there. LRCA does **not** modify SMS; killed set is not filtered. §4.7 H4 numbers use best calibrated combination (`ood_band = 0.02, tolerance_multiplier = 3.0, repeats = 20`); default-threshold results retained in `lrca_60cell_v3.json` as control. Interface with §3.6 risk profile: L0 prescan ↔ §C.2 dual-LLM double-blind; L1 tolerance ↔ N = 20 repetition subprocess; L2 OOD ↔ input-distribution definition; L3 statistical baseline ↔ pre-check protocol.

C.5 C.4.1 Pilot calibration

A 2026 Q3 operator-level pilot ran 12 PUTs × 37 operators (12 `is_key=True` at K=20, 25 at K=10; 470 trials total). Pilot precursor quantities: R_sem (V1-V6 ∧ operator_match pass rate), D_impl (median pairwise AST + literal + identifier Jaccard distance among confirmed mutants), R_kill (proportion of confirmed mutants killed by AVP under that PUT's primary MP). Aggregate (N=37): R_sem median 0.50 mean 0.468 (0/37 with R_sem=0); D_impl median 0.42 mean 0.392 (1/37 with D_impl ≈ 0); R_kill median 0.00 mean 0.189. All 37 operators produced ≥ 1 V1-V6 passing mutant (verifying H1).

Key R_sem / R_kill decoupling pattern. R_sem ≥ 0.9 ∧ R_kill = 0 combinations all on HP / TF / OS operators in c / d classes (e.g., c1_CE1 noise 1e-4→1e-1 with R_sem 1.00, R_kill 0.00; c3_HP1 relu→tanh, c3_TF1 max_iter 1000→5, d1_HP1 MLP α, d3_HP1 LR C). R_sem-high ∧ R_kill = 1 combinations all on CE / OS operators in a / b classes (a2_CE1 LU det, a2_OS1 prod→sum, b1_OS1 α/β swap, d1_TF1 label flip). This is empirical evidence for §5.2 R_sem / R_kill decoupling at cell level.

D D. Statistical Analysis Details

D.1 D.1 SMS variants and construction lemmas

SMS_unfiltered (reviewer comparison metric):

$$\mathrm{SMS}^{\mathrm{unfiltered}}_{i,k,j} := \frac{|\mathrm{killed}_{i,k,j}|}{|\mathrm{mut}_j(S_i)| - |\mathrm{equiv}_{i,k,j}|}$$

(Same form as the primary metric; here `killed` does not distinguish C1 from C2–C5.)

The appendix of the reproducibility package provides cell-by-cell difference between SMS_unfiltered and SMS; if relative difference < 5%, this confirms LRCA does not affect robustness of primary conclusions.

Primary-table construction.

Table 7. SMS variants and construction lemmas.

RQ	Primary statistics	Reporting format
RQ3	inst_rate, equiv_rate, C1_share, survive_rate	60-cell heatmap + 4-class marginals
RQ4	aligned-SMS vs cross-SMS; sparse ∘ vs dense ••equiv_rate	Cliff's delta + odds ratio + 95% bootstrap CI
RQ5	ΔSMS_c (c ∈ {A,B,C,D}); CV(ΔSMS)	sign test (df=3) + descriptive forest plot
RQ5 (desc.)	Spearman ρ + Kendall τ for SMS vs PC	scatter + dual-metric ranking comparison

Multiple comparisons. Cell-level claims across 60 cells use Benjamini-Hochberg FDR, $\alpha_{\mathrm{FDR}} = 0.05$. N = 20 bootstrap intervals: 1,000-iteration 95% CI (B = 10,000 for the headline H2 delta CI).

D.2 D.2 Cliff's delta cutoff sensitivity and logit-transformation invariance

H4 cutoff sensitivity. Dense grid scan (cutoff ∈ {0.05, 0.10, ..., 0.50}, step 0.05) on v4 data (`scripts/h5_sensitivity.py`, `data/results/h5_sensitivity_v4.json`):

Table 8. Cliff's delta cutoff sensitivity and logit-transformation invariance.

cutoff	h5_cells_pass	h5_pass_ratio
0.05	12 / 60	20.0%
0.10	12 / 60	20.0%
0.15	12 / 60	20.0%
0.20 (paper)	**12 / 60**	**20.0%**
0.25	12 / 60	20.0%
0.30	12 / 60	20.0%
0.35	12 / 60	20.0%

0.40	12 / 60	20.0%
0.45	13 / 60	21.7%
0.50	13 / 60	21.7%

Conclusion. H4 verdict (not met) is intrinsic data property, independent of cutoff choice. v4 suspect_share distribution is severely bimodal: median 1.0, mean 0.79; 48 cross cells at suspect_share ≈ 1, 12 aligned cells in low end; the [0.20, 0.80] interior is nearly empty. Specific value 0.20 is **not load-bearing**.

Logit-transformation invariance. Cliff's delta is rank-based (function of $U/(n_1 \cdot n_2)$), mathematically invariant under any strictly monotone transformation (Romano 2006). Logit is strictly monotone on (0, 1), so $\delta_{\text{logit}} \equiv \delta_{\text{raw}}$ is a construction result. **Not additional robustness evidence**; H2 robustness threats come from §4.2 zero-mass dominance, not metric-scale choice.

D.3 D.3 Power analysis, plug-in and stipulated-alternative

Plug-in bootstrap. With-replacement sample from observed aligned (n=12) and cross (n=48) v4 SMS pools; N_sim = 5,000, seed = 42 (`scripts/compute_rq2_power.py`, `rq2_power_v4.json`):

Table 9. Power analysis, plug-in and stipulated-alternative.

Threshold	Interpretation	Power
delta > 0.000	Any-effect	**0.997**
delta > 0.147	Small	0.966
delta > 0.330	Medium	0.759
delta > 0.474	**Large (H2)**	**0.423**

Sample-size sweep ($n_{\text{aligned}} \in \{6, 12, \ldots, 60\}$, $n_{\text{cross}} = 4 \times n_{\text{aligned}}$): power for delta > 0 reaches 0.974 at n=6, 0.996 at 12, plateaus thereafter. RQ4 primary analysis has very low sample-size requirement; the bottleneck is the effect-size boundary itself, not sampling noise. Even at $\delta_{\text{truth}} \approx 0.474$, sample has only ~42% chance of detecting under plug-in; but this does **not** mean "insufficient power is the cause of H2 not being met", observed $\delta_{v3} = 0.323 < 0.474$, observed $\delta_{v4} = 0.314 < 0.474$; increasing sample only narrows CI.

Stipulated-alternative. Implemented in mixture-weight (`scripts/compute_rq2_power_stipulated.py`, N_sim = 2,000): SMS has heavy ties at zero (45/60 = 0), raw shifting is discontinuous (any $\varepsilon > 0$ jumps delta from 0.314 to 0.74). Mixture: aligned' = (probability w) sample from (observed_aligned + 0.001) + (probability 1−w) sample from observed_aligned, calibrate w so E[delta] ≈ 0.474. Calibrated w = 0.094, realised E[delta] = 0.4746.

Table 10. Power analysis, plug-in and stipulated-alternative.

Stipulated truth	Criterion	Power
0.474	$\hat{\delta} \geq 0.474$ (point-estimate)	**0.491**
0.474	95% CI lower > 0 (any-effect)	0.868

Even at $\delta_{\text{truth}} = 0.474$, the (12, 48) design returns $\hat{\delta} \geq 0.474$ in only ~49% of replications, this *supports* the §4.3 framing that "not met" is point-estimate, not effect-size.

D.4 D.4 Friedman per-class, discussion of small-N concordance

Per-class Friedman (3 PUTs × 5 MPs) with Bonferroni × 4: a chi^2=4.00 raw 0.406 adj 1.000, W=0.333; b chi^2=10.78 raw **0.029** adj **0.116**, W=**0.898**; c chi^2=4.00 raw 0.406 adj 1.000, W=0.333; d chi^2=5.00 raw 0.287 adj 1.000, W=0.417. After correction, no per-class result remains significant at family-wise alpha = 0.05. Even b-class W = 0.898 (Cohen 1988 large concordance) is insufficient at N = 3 per class. **H3 verdict rests on §4.5 sign test**; per-class Friedman is sensitivity / descriptive only. Friedman main effect on full 60 cells (chi^2 = 15.30, p = 0.0041) confirms MP rank differences but is logically independent from H3 cross-class consistency.

Mixed-effects unavailability. Primary `sms ~ C(class) + C(operator) + C(class):C(operator) + (1 | put)` returned Singular (insufficient column rank for class × operator interaction; N=60 too small for 11-d fixed-effects + 12 PUT random intercepts). Fallback `sms ~ C(class) + C(operator) + (1 | put)` had PUT random-intercept variance hit boundary 0 (degenerates to OLS). Fallback class p-values (descriptive only): b vs a 0.275 / c vs a 0.892 / d vs a 0.991. RQ5 conclusion shifts to four-piece presentation (class means + sign test + Friedman + forest plot).

D.5 D.5 Pattern-coverage operationalisation

Per PUT, (MP_k, R_outcome ∈ {True, False}) binary-tuple coverage: 5 MPs × 2 outcomes = 10 cells, PC = #triggered / 10. Simplest baseline (per-PUT granularity, not distinguishing mutants). PC range over 12 PUTs: [0.500, 1.000], mean 0.733 (`paper_numbers_v4.json`). Pairing per PUT: Spearman rho = 0.163 (p = 0.613); Kendall tau = 0.136 (p = 0.568).

Power caveat. At n = 12, Spearman 95% CI ≈ [-0.5, +0.6] (Fisher-z approximation); cannot distinguish "zero correlation" from "moderate positive" or "moderate negative". p = 0.61 / 0.57 means "no correlation detected at n = 12", not "correlation does not exist".

Within-class descriptive. b2 PC = 1.0 with mean SMS = 0.067; b1 PC = 0.7 with mean SMS = 0.20. c3 PC = 1.0 with mean SMS = 0.14; c1 / c2 PC = 0.7-0.8 with mean SMS = 0.0. Within the same class, higher-PC PUT has lower SMS, contradicts naive "more PC kills more mutants" assumption. This pattern is consistent with orthogonality, but n = 12 cannot confirm it. A confirmatory study would need n ≥ 30 and a PC definition that incorporates the mutant dimension.

E E. Stakeholder Deployment Considerations

E.1 E.1 Air-gap incompatibility, full justification

The §5.4 workflow depends on external LLM API calls (Claude / GPT / DeepSeek), incompatible with most regulated air-gapped V&V workflows:

Table 11. Air-gap incompatibility, full justification.

Domain	Standard	Air-gap requirement
Nuclear	IEC 60880	Air-gapped build for safety-critical software

Aerospace	DO-178C	Tool-qualified offline build for DAL-A/B; LLM API outside qualified-tool envelope
Medical	IEC 62304	Class C software lifecycle requires offline reproducible build
Automotive	ISO 26262	ASIL-D requires offline tool chain; cloud LLM non-compliant

Mitigation paths. (a) Self-hosted open-weight LLMs (Llama / DeepSeek local inference), capability gap vs API-served frontier models; quantitative impact on SMS untested. (b) Offline-cached pre-generated mutant pools with commit-hash + raw-response signature locking, adopters reproduce a published pool offline (this paper's package supports this), but generating new pools requires external LLM access. SMS for air-gapped industrial deployment is a future research direction.

E.2 E.2 Quarterly batch audit workflow

The original per-PR GitHub Actions YAML was removed because its PR-CI threshold 0.10 contradicted the Appendix E.3 audit threshold 0.20 and propagated stakeholder-facing CI assumptions into adopters' pipelines.

Replacement: quarterly batch audit:

Table 12. Quarterly batch audit workflow.

Step	Description	Cost
1	Offline mutant-pool generation (per PUT 24-30 × 12 PUTs ≈ 300)	LLM API $5-15; ~30 min
2	`sms_campaign.py --track 2`	4-core laptop ~10-20 min
3	`run_lrca.py`	< 1 min
4	MR-design team review of MRs with aligned-cell SMS significantly below historical median (0.275)	1-2 h meeting

Quarterly: API ~$10 + automation 30 min + manual 1-2 h ≈ 0.5 person-day. Affordable. Per-PR gating not recommended (LLM latency 5-30 s + cost + non-determinism + air-gap incompatibility); adopters needing per-PR should use coverage / unit-test gates. SMS does not replace domain-expert MR physical-reasonableness judgement nor product-requirements review.

E.3 E.3 V&V documentation, conceptual complementarity with ASME V&V 20-2009

This appendix subsection makes **no normative claim** toward NRC, FDA, or ISO 26262 review teams. Within single-output kernels scope, no traceable mapping exists to current IEC 60880 / ISO 26262 / DO-178C / ASME V&V 20-2009 normative bodies.

V&V documentation for scientific computing software (per ASME V&V 20-2009 §3 and similar guides) requires quantifiable test-adequacy evidence; current documentation often relies on code coverage + MR lists + SME signatures. SMS may appear as **research-grade supplementary evidence** alongside coverage and MR lists: (a) aligned-cell SMS per critical PUT; (b) LRCA three-layer diagnosis (C1 readout / C2-C5 attribution); (c) 60-cell matrix visualisation. Reviewers can independently run `REPRODUCIBILITY.md`.

Original threshold recommendations (aligned-cell SMS ≥ 0.20 / 0.30 + C1_share ≤ 0.20) were removed: no normative backing in IEC / ISO / ASME; misreads as enforce-ready. **SMS reported as descriptive supplementary evidence**, with adequacy judgements left to V&V reviewers case-by-case.

Conceptually complementary to ASME V&V 20-2009 §3 code verification (numerical-solver correctness) vs SMS (MR-set fault-detection adequacy). Substantial scale gap to multi-module CFD codes; incorporation into V&V standards body would require large-scale empirics + multi-year dialogue with ASME V&V 20 / IEEE 1012 / IEC 60880 committees. **No advocacy that SMS enter any normative certification system in 2027.** SMS does not replace SME signatures, FMECA, PIRT, system-level V&V, or ASME V&V 20 §3 numerical-solver verification, it is one link in the evidence chain, not a single-point qualification.

F F. Threats to Validity, Detailed Mitigation

F.1 F.1 Internal threats

Table 13. Internal threats.

ID	Threat	Mitigation	Residual
R1	LLM generation reproducibility	§C.2 reproducibility parameters in package; dual-LLM cross-source + 20% manual; reproducibility rests on archived raw responses and frozen mutant pools (no user-facing seed)	LLM training data may be updated post-submission
R2	Probabilistic equiv (K_eq=1000)	§2.3 declares probabilistic approximation; Hoeffding-style false-equiv bound	K_eq sweep $\in$ {500,1000,2000} deferred to future work
R3	Shared-infrastructure dependency	AVP version pinned to the upstream commit hash; package embeds AVP source; Li et al. (arXiv) is the infrastructure technical report	None within scope

R4	LRCA multi-label boundary	§A.2 priority C5 > C4 > C3 > C2 > C1; multi-label co-occurrence table in package; root_cause is "likely root cause" not definitive	C2-C5 may multi-trigger on B/D classes
R9	Mutant-pool size	Pool expanded to mean 17.4: delta moved 0.321 → 0.323, CI narrowed; **effect size is intrinsic ceiling, not pool dilution**	Some PUTs < 30 (cache-bound)
R10	LLM non-determinism	Multi-turn de-dup; K=10/20 repetitions; raw prompt+response committed for direct reuse	Claude Opus subscription lacks seed control
R13	**Protocol-implementation gap v3 vs v4**	v4 used 3 providers (Claude 101 / GPT 98 / DeepSeek 99) without dual-blind; v4 C1_share 0.209 > v3 0.164 weakly opposes "v4 quality decline"	Complete separation requires a follow-up dual-blind rerun on the v4 grid

R13 is the protocol-asymmetry threat (dual-blind vs V1-V4 only) on the v3 → v4 source-axis contrast ($\Delta\delta = -0.009$).

F.2 F.2 External / construct / conclusion threats

Table 14. External / construct / conclusion threats.

ID	Threat	Class	Mitigation
R5	12-PUT representativeness	External	follows the shared site selection; open class principle (cls extensible); scaling to MD/QC/CFD
R6	Cross-class statistical power	Conclusion	H3 framed exploratory; mixed-effects unavailable (Singular at N=60); shifted to class means + sign test + Friedman + forest plot

R7	LLM homogeneity bias	Construct	§C.2 three-LLM rotation + PUT-class subdivision + 20% manual sampling; third-party LLM family validation deferred
R8	LLM-source distributional shift	External	Hit distribution DeepSeek 11/15, Claude 4/15, GPT 0/15; §3.5 argument is categorical AST-locality, not hit frequency; HP/SI/TF zero-overlap across all 3 sources unaffected
R12	HOM equivalence	External	Tool-unreachability claim confined to first-order syntactic tools; HOM empirical testing listed as future work

Conclusion threats: multiple comparisons handled by BH-FDR at $\alpha_{\text{FDR}} = 0.05$ (§D.1); N=20 stability handled by 1,000-iteration bootstrap CI (§D.3). Construct threat, SMS measures epistemological semantic detection, not production engineering value (out of scope here).

G G. SMS-to-MS Degeneration Theorem, Full Proof

G.1 G.1 Notation cross-reference

PUT S_i, mutation operator family mut (syntactic or semantic), mutant set $\text{mut}(S_i)$, MP set, equivalence tolerance ε_{eq}, equivalence sample size K_{eq}, AVP tolerance ε_{AVP}. Three-state decomposition: $\text{mut}(S) = \text{killed} \cup \text{equiv} \cup \text{survive}$ (disjoint). SMS formula:

$$\text{SMS}_{i,k,j} = \frac{|\text{killed}_{i,k,j}|}{|\text{mut}_j(S_i)| - |\text{equiv}_{i,k,j}|}$$

G.2 G.2 Degenerate-limit definition

The degenerate limit $L = L_{\text{equiv}} \wedge L_{\text{killed}} \wedge L_{\text{mut}}$ consists of three **joint conditions** (paired axes), each controlling one layer of the SMS formula. L1-L6 are not 6 independent axes; the pairing reflects dependency structure across the SMS formula layers.

Table 15. Degenerate-limit definition.

Joint condition	Component axes	Pairing rationale
L_{equiv} (Lemma 9.1)	L1: $\varepsilon_{\text{eq}} \to 0$; L2: $K_{\text{eq}} \to \infty$	L1 alone: equiv stays probabilistic (K_{eq} cannot cover D_S). L2 alone: bitwise equality diluted by ε_{eq}. Both simultaneously degenerate equiv to classical behavioural equivalence outside a D_S-measure-zero set.
L_{killed} (Lemma 9.2)	L3: $\varepsilon^{k}_{\text{AVP}} \to 0$; L4: MP = {$\text{MP}_{\text{eq}}$}, R(y,y') ≡ y = y'	L3 alone: $\varepsilon_{\text{AVP}} \to 0$ still allows non-trivial MP (monotonicity, convergence). L4 alone: equality still carries ε_{AVP} tolerance. Both simultaneously degenerate killed to classical difference detection.
L_{mut} (Lemma 9.3)	L5: $\text{mut}_j \to$ rule-based syntactic (Mothra AOR/ROR/SDL/CRP); L6: cls(I) ⊆ {imperative deterministic}	L5 alone: syntactic ops on probabilistic/ML may still trigger domain-semantic subsets. L6 alone: imperative programs still mutable by OS/HP/TF/SI. Both simultaneously degenerate mut(S) to Jia & Harman syntactic mutant set.

G.3 G.3 Lemmas, Three-state decomposition degenerates under L

Lemma 9.1 (equiv degeneration; measure-zero qualification). Under L_{equiv} (L1 ∧ L2), semantic-class equivalence (E1 ∧ E2) degenerates to classical behavioural equivalence **almost everywhere** w.r.t. measure D_S.

Proof. E1 (type consistency) holds trivially under $\varepsilon_{\text{eq}} \to 0$ (L6 makes imperative output spaces scalar/vector with static types). E2 ($|S_i(x) - s'(x)| < \varepsilon_{\text{eq}}$ for K_{eq} samples) under L1 ($\varepsilon_{\text{eq}} \to 0$) ∧ L2 ($K_{\text{eq}} \to \infty$ with measure-equivalent sampling) is almost-everywhere equivalent to $\forall x \in D_S \setminus N : S_i(x) = s'(x)$, where N is a D_S-measure-zero set (continuous D_S: floating-point pathological points / NaN propagation; discrete D_S: $N = \emptyset$). This matches Jia and Harman [1] §3 classical equivalent-mutant definition under measure-zero equivalence classes.

Lemma 9.2 (killed degeneration). Under L3 ∧ L4, killed degenerates to classical difference detection.

Proof. L4 restricts MP to {MP_eq} with R(y, y') ≡ y = y'. For mr = (r, R) and mutant s', the MP_{eq} violation condition $\exists x : S_i(x) \neq s'(r(x))$ under L3 ($\varepsilon_{\text{AVP}} \to 0$) becomes exact inequality. With r = id, violation is $S_i(x) \neq s'(x)$, classical difference detection. With r ≠ id, MP_{eq} still requires $S_i(x) = s'(r(x))$ as reference oracle from original program; no new state classifications introduced.

Lemma 9.3 (mut degeneration). Under L5 ∧ L6, mut_j(S_i) degenerates to the syntactic mutant set of Jia and Harman [1].

Proof. L5 switches mut_j to rule-based syntactic operators (AOR, ROR, SDL, CRP, UOI, standard Mothra/Proteum sets); L6 restricts PUTs to imperative deterministic programs, excluding triggering conditions for semantic operators on probabilistic/ML programs. Under this configuration, $\mathrm{mut}_j(S_i)$ is the literature-defined syntactic mutant set, independent of domain semantics.

G.4 G.4 Theorem 9.1, Detailed proof

Theorem 9.1 (SMS → MS degeneration). In the degenerate limit $L = L_{\mathrm{equiv}} \wedge L_{\mathrm{killed}} \wedge L_{\mathrm{mut}}$, almost everywhere w.r.t. D_S,

$$\mathrm{SMS}_{i,k,j} \xrightarrow{L} \mathrm{MS}_{i,j} := \frac{|\mathrm{killed}^{\mathrm{classic}}_{i,j}|}{|\mathrm{mut}^{\mathrm{syntax}}_j(S_i)| - |\mathrm{equiv}^{\mathrm{classic}}_{i,j}|}$$

Proof. Combine Lemmas 9.1-9.3:

- Numerator: under L3 ∧ L4, $\mathrm{killed}_{i,k,j} \to \mathrm{killed}^{\mathrm{classic}}_{i,j}$ (Lemma 9.2); L4 makes $\mathrm{MR}_{i,k}$ trivial over k (only $\mathrm{MP}_{\mathrm{eq}}$ remains), so subscript k degenerates.
- Denominator $|\mathrm{mut}_j(S_i)|$: under L5 ∧ L6 → $|\mathrm{mut}^{\mathrm{syntax}}_j(S_i)|$ (Lemma 9.3).
- Denominator $|\mathrm{equiv}_{i,k,j}|$: under L1 ∧ L2 → $|\mathrm{equiv}^{\mathrm{classic}}_{i,j}|$ (Lemma 9.1; almost-everywhere w.r.t. D_S).

Substituting into the SMS formula gives $\mathrm{MS}_{i,j}$.

Corollary 9.1 (LRCA trivialisation). Under $L = L1 \wedge L2 \wedge L3$, C = {C1, ..., C5} degenerates to {C1}.

Sketch. Each of C2-C5's triggering precondition depends on at least one L_j being violated. When L1 ∧ L2 ∧ L3 hold simultaneously, every non-trivial-space dimension (MP non-triviality, AVP tolerance non-zero, non-empty equiv set, MR-design DOF, class-mapping openness) closes; C2-C5 triggering set becomes empty (read off from §A.2 decision tree). The per-C_k to per-L_j minimum-sufficient mapping depends on §3.11 LRCA-classifier engineering thresholds; we do not claim one-to-one correspondence at formal level, readers can trace through §A.2 + §C.4. Under L, `suspect_share` → 0, LRCA reports only C1, SMS degenerates to single-layer metric consistent with Jia and Harman [1] MS engineering structure.

Empirical consistency. Theorem 9.1 + Corollary 9.1 jointly guarantee any SMS-based empirical conclusion (Cliff's delta §4.3, Friedman chi^2 §4.5, Spearman rho §4.6) is structurally consistent with existing Jia and Harman [1] literature in classical syntactic-mutation scenarios, no metric-level semantic fragmentation.

H H. Adjoint Extension Arm, Subjects, Pools, and Forward Results

This appendix gives the full specification of the confirmatory adjoint extension arm summarised in §3.7 and §4.9 of the main paper. The arm exercises the adjoint invariant ψ_6 on three third-party linear-operator libraries, disjoint from the 12 PUTs and the pre-registered 60-cell. Each library carries a real fixed defect in its adjoint code path as a ground-truth anchor:

- **scipy.sparse.linalg LinearOperator** (linear algebra; BSD-3). Program under test: the scaled-operator adjoint M^{H} for $M = \alpha A$ with complex scale α. Real anchor: scipy issue #8900 / PR #8962 (the adjoint dropped the conjugate on α, returning αA^{H} instead of $\bar{\alpha} A^{\mathrm{H}}$).
- **pylops.signalprocessing.FFT** with `real=True` (signal processing; BSD-3). Program under test: the real-FFT adjoint `_rmatvec`; the adjoint identity is the library's own *dot-test*. Real anchor: pylops issue #268 / PR #292 (commit `bd3a919`), a wrong scaling of the middle/Nyquist frequency bin that made the adjoint inconsistent with the forward.

- **jax** `linear_transpose` / `vjp` (automatic differentiation; Apache-2.0). Program under test: the autodiff-derived transpose M^{T} of a linear operator containing an inverse real FFT, $M(x) = \mathrm{irfft}(\mathrm{rfft}(x) \cdot w)$. Real anchor: jax issue #6223 / PR #6232 (commit `112ae41`), an `irfft` transpose rule that reversed order with an off-by-one index, so `vjp` silently produced a non-adjoint transpose.

For each subject the MR families are adj.self (self-adjointness) and adj.dual (scaled/dual-operator transpose). The mutant pool spans adjoint-breaking fault mechanisms (drop-conjugate, sign flip, scale and phase errors, transpose-without-conjugation, Nyquist mis-scaling), with the real fixed defect included among them, plus semantically equivalent mutants (scalar-multiplication reorder, conjugation of a real adjoint output) that a strong MR must *not* kill. Because the pre-fix releases are uninstallable on the host platform (no `arm64` wheel), each real defect is reproduced in extracted-diff form on the current release, mirroring the boundary study; $\varepsilon_{\mathrm{tol}}$ is an explicit factor throughout.

Table 16. Adjoint extension arm, forward kill.

Substrate	Operator class (real anchor)	Active killed / total	Equiv. survived	Forward SMS
scipy LinearOperator	linear algebra (#8900)	5 / 5	2 / 2	**1.00**
pylops FFT real	signal processing (#292)	4 / 4	2 / 2	**1.00**
jax vjp / transpose	automatic differentiation (#6223)	4 / 4	2 / 2	**1.00**

Across all three substrates the correct program survives, every non-equivalent (active) mutant is killed, including the three real fixed defects, and no semantically equivalent mutant is killed, so the forward SMS is 1.00 with zero false positives on the equivalent set. The scipy #8900 defect is shared with the §4.8 boundary study (adj.self case) and is not counted as independent corroboration twice; the pylops #292 and jax #6223 defects are unique to this arm. The pools are hand-constructed and low-cardinality, so the arm demonstrates the duality forward rather than measuring a high-adequacy MR set on a discovered fault population (main-paper threat R14).

I I. Result-Level Real-Defect Evidence Ledger

This appendix records only the result-level facts used by the main manuscript's real-defect evidence slice. It deliberately excludes candidate mining, rejected cases, repository tooling, and benchmark-release narrative. The purpose is auditability for the semantic-mutation argument: kill-rate, semantic-stratum alignment, and real-defect detection are related but distinct constructs.

Table 17. Result-level real-defect evidence ledger.

Audit item	Ledger content used in the manuscript	Main-text role
Verified stratum counts	34 verified MR-detectable defects across five meta-patterns: m_{mono} 10, m_{conv} 9, m_{inv} 6, m_{adj} 5, and m_{rev} 4	Shows that the declared semantic strata are not only toy labels

Mutation-phase aggregate	T1–B1 mean difference +0.101, Holm-adjusted $p = 0.046$, Cliff's $\delta = +0.247$; T1–A1 not significant; B1–B2 not significant	Supports a modest, qualified mutation-score advantage rather than a dominance claim
Real-defect face	T1 detects 30/30 tabled real defects; B1 has nonzero detection in 4/30, B2 in 6/30; A1-a loses 17/30, A1-b loses 16/30, and both lose 10/30	Separates real-defect detection from aggregate mutant kill-rate
Non-nesting cases	A-LAPACK-004, A-OPENBLAS-001, B-POCKETFFT-002, and E-ORDINARYDIFFEQ-001	Supports the semantic-effect fiber view rather than a single scalar hierarchy

The ledger is intentionally a result table, not a corpus or benchmark description. The main manuscript uses it only to support construct separation: a relation family can score similarly on aggregate mutant kill-rate while differing sharply on real-defect detection, and different relation families can observe non-nested semantic-effect fibers.

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